

Twilio Field Guide

Most Twilio failures are a setting, not a bug. A number still points at the demo TwiML URL, a campaign never finished registering, a webhook has no fallback. Every script here is read only.

40 fixes, each one a written note with diagrams and a small Python and Node.js script that finds the problem and repairs it. All 40 have a script in the public repo.

Before you run anything. Every script starts in a dry run: it reports what it would change and writes nothing. Read that output first, then set `DRY_RUN=false` when you are satisfied it has understood your data.

Scripts: github.com/allanninal/twilio-fixes

Guides: allanninal.dev/twilio

All 14 repos: github.com/allanninal

The fixes

1 an A2P brand stuck at FAILED blocks every campaign under it

Campaign creation keeps getting rejected and every US message comes back 30034, so the team keeps looking at the campaign. The campaign is not the problem: the brand above it is FAILED, nothing can attach to a failed brand, and the reason it failed has been sitting in `errors[]` on the brand resource since the day it was reviewed.

[twilio_a2p_brand_audit.py](#)

<https://www.allanninal.dev/twilio/a2p-brand-registration-failed/>

<https://github.com/allanninal/twilio-fixes/tree/main/a2p-brand-registration-failed>

2 an a2p campaign parked at IN_PROGRESS is not a live campaign

Registration was submitted, the console went quiet, and three weeks later the launch shipped on the assumption that quiet meant done. It did not. `campaign_status` still reads `IN_PROGRESS`, `campaign_id` is still null, and every US message is coming back 30034. Nothing is broken. It simply was never approved.

[twilio_a2p_campaign_wait_audit.py](#)

<https://www.allanninal.dev/twilio/a2p-campaign-stuck-in-progress/>

<https://github.com/allanninal/twilio-fixes/tree/main/a2p-campaign-stuck-in-progress>

3 **a2p campaign is FAILED and errors[] names the rejected field**

The campaign was rejected in March. Somebody resubmitted the same description in April and it was rejected again. The reason was in the response both times: errors[] on the campaign carries a code, a sentence of English and the exact fields that triggered it — and the dashboard the team built reads campaign_status and stops there.

[twilio_a2p_campaign_vetting_audit.py](#)

<https://www.allanninal.dev/twilio/a2p-campaign-vetting-failed/>

<https://github.com/allanninal/twilio-fixes/tree/main/a2p-campaign-vetting-failed>

4 **no API keys exist, so the auth token is the credential**

Every other note in this section tells you to run its script with an API Key that has read access rather than the account auth token. This is the one that explains why, and how to find out whether anything you run is still holding the token. Nothing is failing right now. That is the shape of the problem: it costs you nothing until the day you have to rotate, and then it costs you everything at once.

[twilio_credential_audit.py](#)

<https://www.allanninal.dev/twilio/auth-token-used-instead-of-api-key/>

<https://github.com/allanninal/twilio-fixes/tree/main/auth-token-used-instead-of-api-key>

5 **error 21617: the rendered message body exceeds 1600 chars**

The template is fine. It has been fine for a year. Then one customer with a long company name, three line items and a German address renders past sixteen hundred characters, Twilio refuses the request with 21617, and that customer never receives the message. Their Message SID is not in your logs because there is no Message SID: the send was rejected before a resource was created, so it appears nowhere in the Messages list at all.

[twilio_body_length_audit.py](#)

<https://www.allanninal.dev/twilio/body-exceeds-1600-chars-21617/>

<https://github.com/allanninal/twilio-fixes/tree/main/body-exceeds-1600-chars-21617>

6 **carrier filtering drops your SMS silently with error 30007**

The API returned 201. The Message SID is in your logs. The status walked queued, sent, and then undelivered with error_code 30007, and the recipient saw nothing at all. You were billed for the attempt. No HTTP request failed, no webhook errored, nothing appeared in your exception tracker — a carrier, or Twilio itself, read the message and dropped it.

[twilio_filtered_messages_audit.py](#)

<https://www.allanninal.dev/twilio/carrier-filtered-messages-30007/>

<https://github.com/allanninal/twilio-fixes/tree/main/carrier-filtered-messages-30007>

7 **recycled numbers send OTPs to whoever owns them now**

Every message says delivered. The password reset code, the appointment reminder, the balance alert — all of them accepted by the carrier, all of them handed to a handset. Just not the handset you think. The number was disconnected eleven weeks ago and reissued to somebody who has never heard of you, and your contact table has not noticed.

[twilio_deactivations_audit.py](#)

<https://www.allanninal.dev/twilio/deactivated-number-recycling/>

<https://github.com/allanninal/twilio-fixes/tree/main/deactivated-number-recycling>

8 Dial rejected with 13214 on a passed-through caller ID

Call forwarding works. It has worked all week. Then a handful of calls fail, and they fail without a pattern anyone can see — not one number, not one hour, not one destination. What they have in common is invisible from the outside: the inbound leg arrived carrying a caller ID the terminating carrier will not accept, your <Dial> passed it through unchanged, and Twilio logged 13214 Dial: Invalid callerId value against a call nobody was watching.

[twilio_dial_caller_id_audit.py](#)

<https://www.allanninal.dev/twilio/dial-invalid-caller-id-13214/>

<https://github.com/allanninal/twilio-fixes/tree/main/dial-invalid-caller-id-13214>

9 Fraud Guard blocked the prefix, so real users get 60410

Support has four tickets from one country, all saying the same thing: the code never arrives. Your logs show 60410, verification delivery attempt blocked. Nothing in the account changed, no carrier is down, and the numbers are ordinary mobiles. Fraud Guard is doing exactly what you asked it to do — it found pumping-shaped traffic on a number prefix and stopped sending there for twelve hours — and your real users happen to live on that prefix too.

[twilio_fraud_guard_block_audit.py](#)

<https://www.allanninal.dev/twilio/fraud-guard-blocking-prefix/>

<https://github.com/allanninal/twilio-fixes/tree/main/fraud-guard-blocking-prefix>

10 a voice-only From number fails every SMS with error 21606

The number works. People call it, the IVR answers, it has been on the account for two years. Then a new notification job starts sending from it and every single message is rejected with 21606: 'From' number is not a valid message-capable Twilio number for this account. Both halves of that sentence are load-bearing, and only one of them is about SMS.

[twilio_from_number_capability_audit.py](#)

<https://www.allanninal.dev/twilio/from-number-not-sms-capable/>

<https://github.com/allanninal/twilio-fixes/tree/main/from-number-not-sms-capable>

11 phone numbers with no traffic still bill every month

Nothing is broken. The invoice creeps up while message volume stays flat, and when somebody finally asks what the forty-one numbers on the account are for, the honest answer is that nobody knows. There is no error to search for, no alert to acknowledge — just a line item that has been quietly compounding since the last time anyone looked.

[twilio_idle_numbers_audit.py](#)

<https://www.allanninal.dev/twilio/idle-phone-numbers-billed/>

<https://github.com/allanninal/twilio-fixes/tree/main/idle-phone-numbers-billed>

12 inbound SMS disappears into a number with no sms_url

Outbound works perfectly. Replies do not arrive. There is no 4xx, no entry in the Debugger, no request in your access log — the inbound message is accepted by Twilio, matched to a number, and then delivered to nowhere. The STOP replies vanish the same way, which is the part that eventually costs money.

[twilio_inbound_route_audit.py](#)

<https://www.allanninal.dev/twilio/inbound-webhook-black-hole/>

<https://github.com/allanninal/twilio-fixes/tree/main/inbound-webhook-black-hole>

13 SMS to a landline fails with 30006 and retrying never helps

The same twelve numbers fail every night. `error_code` 30006, status undelivered, and a retry scheduled by a queue that assumes failures are temporary. They are not. Those numbers are desk phones, and no amount of retrying will make a desk phone receive an SMS — but you are billed for each attempt, and the customer is on your list as unreachable rather than as unreachable-by-this-channel.

[twilio_landline_audit.py](#)

<https://www.allanninal.dev/twilio/landline-destination-30006/>

<https://github.com/allanninal/twilio-fixes/tree/main/landline-destination-30006>

14 messages stay queued or accepted and never reach a final state

The send succeeded four hours ago. status is still queued. `error_code` is null, `date_sent` is null, and the status callback has never fired because nothing has happened to report. Nobody has been paged, because nothing has failed yet — and in six hours these will start failing with 30001 or expiring with 30036, long after the passcode they carry stopped being useful.

[twilio_stuck_messages_audit.py](#)

<https://www.allanninal.dev/twilio/messages-stuck-queued-or-accepted/>

<https://github.com/allanninal/twilio-fixes/tree/main/messages-stuck-queued-or-accepted>

15 queue overflow 30001: a send loop outruns one long code

The nightly job dispatched forty thousand messages in about eleven minutes, the way it always has. This time six thousand of them came back with `error_code` 30001, some of the rest were rejected at request time with 21611, and the ones that survived arrived the following afternoon. Nothing in the code changed. The list got longer, and a single long code can only send about one message a second.

[twilio_queue_overflow_audit.py](#)

<https://www.allanninal.dev/twilio/messaging-queue-overflow-30001/>

<https://github.com/allanninal/twilio-fixes/tree/main/messaging-queue-overflow-30001>

16 an empty sender pool fails every send with error 21704

The Messaging Service was created by a setup script in March. It has a friendly name, a SID your application has been passing on every send since, an inbound webhook, a status callback — and nothing in the sender pool. Every `Messages.create` against it comes back 21704, “The Messaging Service contains no phone numbers”, before a carrier is ever involved.

[twilio_sender_pool_audit.py](#)

<https://www.allanninal.dev/twilio/messaging-service-empty-sender-pool/>

<https://github.com/allanninal/twilio-fixes/tree/main/messaging-service-empty-sender-pool>

17 no status callback means delivery failures never reach you

Your dashboard says every message sent. Support says customers never got them. Both are true: `Messages.create` returned queued, your code wrote sent, and the undelivered that arrived ninety seconds later went to a status callback that was never configured. The 21610s, the 30007s and the 30034s exist, in Twilio's logs, where nothing you own is looking.

[twilio_delivery_observability_audit.py](#)

<https://www.allanninal.dev/twilio/messaging-service-no-status-callback/>

<https://github.com/allanninal/twilio-fixes/tree/main/messaging-service-no-status-callback>

18 **a Messaging Service with no A2P campaign fails US sends**

Staging was cloned from production last quarter and it has worked ever since — until the new tenant went live and every US message came back 30034. The brand is approved. The campaign is verified. Neither of them is attached to this Messaging Service, because A2P registration is per service and nobody registers the second one.

[twilio_a2p_registration_audit.py](#)

<https://www.allanninal.dev/twilio/messaging-service-not-a2p-registered/>

<https://github.com/allanninal/twilio-fixes/tree/main/messaging-service-not-a2p-registered>

19 **no Usage Trigger, so overspend runs with nothing watching**

The invoice is for eleven thousand dollars. Most of it is SMS to a country you do not sell in, sent over a Saturday night by the verification form on your signup page. Nobody was paged, because there was nothing to page: Usage Triggers are the only spend alarm Twilio runs on its own side, and an account is created with none of them.

[twilio_usage_trigger_audit.py](#)

<https://www.allanninal.dev/twilio/no-usage-trigger-configured/>

<https://github.com/allanninal/twilio-fixes/tree/main/no-usage-trigger-configured>

20 **a number with an Application SID ignores its own voice_url**

You changed voice_url on the number this morning. You changed it again in the console an hour later and watched the page save. Calls keep arriving at an endpoint you retired last spring. Nothing is broken and nothing is cached — voice_application_sid is set on that number, and while it is, the field you keep editing is not read at all.

[twilio_number_app_precedence_audit.py](#)

<https://www.allanninal.dev/twilio/number-conflicting-url-and-application-sid/>

<https://github.com/allanninal/twilio-fixes/tree/main/number-conflicting-url-and-application-sid>

21 **sends to recipients who texted STOP bounce with 21610**

Someone replied STOP four months ago. Twilio recorded it, blocked that sender from reaching them, and has been rejecting your sends with 21610 ever since. You were never charged, so nothing showed up on the bill; your send queue treated the rejection as a transient failure and retried; and the only place the whole story exists is the Messages list, which has no filter for it.

[twilio_opt_out_audit.py](#)

<https://www.allanninal.dev/twilio/opted-out-recipients-21610/>

<https://github.com/allanninal/twilio-fixes/tree/main/opted-out-recipients-21610>

22 **a rising share of outbound calls end in status failed**

Nobody can point at an error. Support says calls are not going through; the Debugger has its usual scattering of alerts and none of them is new; the code has not changed. What has changed is a ratio: the share of outbound calls ending in failed rather than completed. A ratio is not an event, so nothing raised it, and nothing will — you have to go and compute it.

[twilio_call_failure_rate_audit.py](#)

<https://www.allanninal.dev/twilio/outbound-call-failure-rate-spike/>

<https://github.com/allanninal/twilio-fixes/tree/main/outbound-call-failure-rate-spike>

23 **outbound messaging is off, so every send fails with 30037**

One tenant stopped receiving messages. Not slowly, not partially — every send from that subaccount comes back with `error_code=30037`, “outbound message not allowed”, while the other nineteen tenants on the same code, the same numbers and the same deploy are entirely unaffected. Nothing changed on your side, which is exactly what makes it hard to look for.

[twilio_outbound_disabled_audit.py](#)

<https://www.allanninal.dev/twilio/outbound-messaging-disabled-30037/>

<https://github.com/allanninal/twilio-fixes/tree/main/outbound-messaging-disabled-30037>

24 **a number with no fallback URL drops the call when yours 500s**

Your webhook was down for ninety seconds during a deploy. Twilio requested it, got a 502, logged an 11200 and hung up on whoever was calling. There was a mitigation for exactly this, it costs one field on the number, and it is empty on every number you own.

[twilio_fallback_audit.py](#)

<https://www.allanninal.dev/twilio/phone-number-missing-fallback-url/>

<https://github.com/allanninal/twilio-fixes/tree/main/phone-number-missing-fallback-url>

25 **a phone number still points at Twilio's demo TwiML**

The number rings. It answers. It plays a cheerful message about Twilio and hangs up. Nothing appears in the Debugger, nothing appears in your logs, and every call in the console is marked completed — because the webhook Twilio fetched answered perfectly. It just was not yours.

[twilio_demo_twiml_audit.py](#)

<https://www.allanninal.dev/twilio/phone-number-still-on-demo-twiml/>

<https://github.com/allanninal/twilio-fixes/tree/main/phone-number-still-on-demo-twiml>

26 **an approved regulatory bundle is counting down to expiry**

Eighteen months of German numbers working perfectly, and then one Tuesday they stop. Nothing was deployed. The bundle that proved your business address to the regulator reached its `valid_until`, flipped out of `twilio-approved` on its own, and the numbers attached to it are now non-compliant. There was a date, it was in the API the whole time, and nothing ever mentioned it.

[twilio_bundle_expiry_audit.py](#)

<https://www.allanninal.dev/twilio/regulatory-bundle-expiring/>

<https://github.com/allanninal/twilio-fixes/tree/main/regulatory-bundle-expiring>

27 **a SIP Domain with no `auth_type` accepts no traffic at all**

The domain exists. It has a name, it has a `voice_url`, it appears in the console alongside the ones that work, and it rejects every call. Not with a 500, not with a TwiML error, not with anything your application can see — the INVITE is refused at authentication, which happens before Twilio ever looks at the URL you configured. The domain looks provisioned because provisioning it and making it able to accept traffic are two different operations.

[twilio_sip_domain_auth_audit.py](#)

<https://www.allanninal.dev/twilio/sip-domain-no-auth-type/>

<https://github.com/allanninal/twilio-fixes/tree/main/sip-domain-no-auth-type>

28 SMS Pumping Protection blocks legitimate OTPs with 30450

The one-time passcodes were arriving. Then, for one country, they stopped: error_code 30450, a few hundred of them, over about twenty minutes. By the time the first support ticket reached anyone the sends had resumed on their own and every dashboard was green again. Nothing in your code changed, nothing in the account changed, and there is nothing left to point at — except a login page where a few hundred people could not get in.

[twilio_pumping_block_audit.py](#)

<https://www.allanninal.dev/twilio/sms-pumping-protection-30450/>

<https://github.com/allanninal/twilio-fixes/tree/main/sms-pumping-protection-30450>

29 years-old API keys are still live with nobody owning them

There are eleven keys on the account. Four have names. One of the unnamed ones was created in March 2023 by a contractor whose email address stopped working two years ago, and nobody can say what it authenticates. It has never expired, because Twilio keys do not. It has no last-used timestamp, because that field does not exist. And deleting it to find out will also invalidate every Access Token it ever signed.

[twilio_api_key_audit.py](#)

<https://www.allanninal.dev/twilio/stale-or-orphaned-api-keys/>

<https://github.com/allanninal/twilio-fixes/tree/main/stale-or-orphaned-api-keys>

30 status callback failures with 11200 leave delivery state blind

Support says the customer never got the text. Your dashboard agrees: the row still reads queued, hours later. Then you open the Twilio Console, paste the Message SID, and it says delivered — forty seconds after you sent it. Nothing was lost. Twilio tried to tell you, your endpoint returned something other than a 2xx, and the update went in the bin along with every other one that day.

[twilio_status_callback_audit.py](#)

<https://www.allanninal.dev/twilio/status-callback-webhook-failing-11200/>

<https://github.com/allanninal/twilio-fixes/tree/main/status-callback-webhook-failing-11200>

31 an unverified toll-free number is blocked, not throttled

Toll-free was the easy option: no brand, no campaign, no EIN, buy the number and send. That stopped being true on 31 January 2024. Unverified toll-free traffic to US and Canadian mobiles is now blocked outright rather than throttled, every message comes back 30032, and you are still billed for the attempts.

[twilio_tollfree_verification_audit.py](#)

<https://www.allanninal.dev/twilio/tollfree-number-not-verified/>

<https://github.com/allanninal/twilio-fixes/tree/main/tollfree-number-not-verified>

32 a trial account rejects multi-segment messages with 30044

“Test” sends. “Your verification code is 481920” sends. The real welcome message, the one with the customer's name and a link and a single cheerful emoji at the end, comes back undelivered with error_code=30044. Everyone's first theory is the link. It is not the link.

[twilio_trial_segment_audit.py](#)

<https://www.allanninal.dev/twilio/trial-account-segment-limit-30044/>

<https://github.com/allanninal/twilio-fixes/tree/main/trial-account-segment-limit-30044>

33 **a SIP trunk with no disaster recovery URL loses every call**

The trunk works. It has worked for a year. Then the PBX reboots, or the firewall rule expires, or the datacentre link flaps for four minutes, and every inbound call in those four minutes is dropped at Twilio — not sent to voicemail, not answered by an apology, dropped. The field that would have caught them is empty, and it has always been empty, because nothing ever asked you to fill it in.

[twilio_trunk_dr_audit.py](#)

<https://www.allanninal.dev/twilio/trunk-missing-disaster-recovery-url/>

<https://github.com/allanninal/twilio-fixes/tree/main/trunk-missing-disaster-recovery-url>

34 **one smart quote triples your segment count and your bill**

Nothing failed. Every message says delivered, every customer got it, and the only thing that changed is the invoice: the SMS line is three times what it was on the same send volume. Somewhere in a template, an edit made in a rich text box replaced a straight apostrophe with a curly one. Every message that template renders now costs three segments instead of one, and there is no error code anywhere in the account to say so.

[twilio_segment_audit.py](#)

<https://www.allanninal.dev/twilio/ucs2-segment-inflation/>

<https://github.com/allanninal/twilio-fixes/tree/main/ucs2-segment-inflation>

35 **Verify conversion collapses in one country: SMS pumping**

The Verify bill for the month is five times last month's. Nothing is failing: no 4xx, no error code, no delivery problem, no support tickets. The sends are being accepted, delivered and charged exactly as designed. The only thing that changed is that in one country almost nobody types the code in any more — and that single number is the difference between a growth spurt and somebody quietly farming your signup form.

[twilio_verify_conversion_audit.py](#)

<https://www.allanninal.dev/twilio/verify-conversion-rate-collapse/>

<https://github.com/allanninal/twilio-fixes/tree/main/verify-conversion-rate-collapse>

36 **a Verify Service with zero rate limits configured**

Verify looks protected. There is a limit on how often one phone number can be sent a code, and you have seen 60212 in the logs, so something is clearly throttling something. Then a scripted signup endpoint sends ten thousand verifications to ten thousand different numbers from a single host, hits no limit at all, and the invoice explains why: the protection you were relying on is keyed on the destination, and the attacker changes the destination every time.

[twilio_verify_rate_limit_audit.py](#)

<https://www.allanninal.dev/twilio/verify-no-rate-limits/>

<https://github.com/allanninal/twilio-fixes/tree/main/verify-no-rate-limits>

37 **Verify sends SMS to a landline: 60205, or just silence**

A small, stubborn fraction of your signups never complete. They are not bots, they do not retry three times and give up, and support cannot reproduce any of it. The numbers look fine: right length, right country, valid E.164. They are landlines. An SMS to a landline either comes back 60205 or, if Lookup is off, disappears into a verification that stays pending until it expires — billed, delivered nowhere.

[twilio_verify_landline_audit.py](#)

<https://www.allanninal.dev/twilio/verify-sms-to-landline/>

<https://github.com/allanninal/twilio-fixes/tree/main/verify-sms-to-landline>

38 **twilio cannot open a TCP connection to your webhook (11205)**

You have the URL open in a browser tab and it works. Curl works. The health check is green. And the Twilio Debugger is filling with 11205 HTTP connection failure for that exact URL, while your access log has no entry for it at all — not a 500, not a 404, nothing. The request never arrived, because the connection was never opened.

[twilio_webhook_timeout_audit.py](#)

<https://www.allanninal.dev/twilio/webhook-connection-timeout-11205/>

<https://github.com/allanninal/twilio-fixes/tree/main/webhook-connection-timeout-11205>

39 **a webhook hostname with no public DNS record fails with 11210**

It worked on the laptop it was written on. It worked in staging. It reached production and every inbound call to that number now produces 11210 HTTP bad host name, because the hostname in the webhook resolves through an /etc/hosts line, an internal zone, or a tunnel that died when someone closed a terminal. Twilio resolves from the public internet and gets nothing back.

[twilio_webhook_dns_audit.py](#)

<https://www.allanninal.dev/twilio/webhook-dns-resolution-failure-11210/>

<https://github.com/allanninal/twilio-fixes/tree/main/webhook-dns-resolution-failure-11210>

40 **an expired webhook certificate fails every request with 11236**

At 14:07 UTC everything was fine. At 14:08 every webhook on one hostname started failing with 11236 Certificate Invalid - Certificate Expired, and it has not stopped since. No deploy went out. No configuration changed. A renewal job stopped working ninety days ago and nobody noticed, because a certificate does not degrade — it works perfectly until the second it does not.

[twilio_webhook_cert_audit.py](#)

<https://www.allanninal.dev/twilio/webhook-tls-certificate-expired-11236/>

<https://github.com/allanninal/twilio-fixes/tree/main/webhook-tls-certificate-expired-11236>
